On The Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

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Abstract

Background: Conserved metabolite pools and substrate availability constrain biochemical pathways before kinetic parameters are known. Analysing these constraints across curated and rule-generated networks requires both reaction stoichiometry and the amounts of substrates consumed at each step. Participants shared by both sides of a reaction can disappear from net stoichiometry while remaining necessary for its execution.

Results: We present the chemical reaction network (CRN) submodule of `SynKit`, an open-source Python toolkit, for constructing biochemical reaction networks and analysing their conserved pools, chemical reaction network theory (CRNT) properties, and executable reaction sequences. `KEGG` pathways, supported `SBML` models, and rule-generated reactions use the same reaction records. Twelve reference networks pass the validation checks and agree with `CRNT4SBML` on eight shared structural quantities. Ten reaction-bearing `BioModels` entries pass supported structural round trips, including a model with 504 species and 827 directed reactions. Four `KEGG` modules illustrate conserved pools and the effects of model boundaries. In glycolysis, removing shared polyphosphate from a reaction changes deficiency and substrate requirements while preserving the net stoichiometric matrix. A 15-step glucose-to-pyruvate sequence executes after currency metabolites are removed but is blocked from glucose alone when they are retained. In a generated formose network, conservation coefficients match carbon counts for all 37 species, but equal carbon inventories differ in their ability to supply the selected tetrose and hexose routes. Benchmarks identify exact conserved-pool enumeration as a practical limit for larger networks.

Conclusions: `SynKit` brings conservation analysis and reaction-by-reaction execution checks to the same biochemical model. Its results help distinguish conserved quantities, unmet substrate requirements, and the assumptions needed for CRNT conclusions. Source code, recorded results, and reproducible analyses accompany the manuscript.

Keywords: Chemical reaction networks, chemical reaction network theory, Petri nets, conserved moieties, stoichiometry, pathway realizability, `SBML`

1 Background

Biochemical pathway analysis increasingly combines curated reactions with computationally generated alternatives. `KEGG`, `Reactome`, and `WikiPathways` provide reaction records for reuse [1, 2, 3], while graph-transformation methods enumerate reactions from molecular seeds and transformation rules [4]. Before assigning kinetic parameters, researchers need to determine which quantities a network conserves, which structural theorems apply, and whether a proposed route can execute from the available substrates. Answering these questions on the same reaction system is useful for checking metabolic modules and screening generated reaction spaces.

The central difficulty is that these analyses require different information. A species graph represents connectivity but can lose the joint requirement of multiple substrates [5]. The net stoichiometric matrix records changes in species amounts but cancels participants present on both sides: with a fixed species inventory, $A + C \to B + C$ and $A \to B$ have identical net columns. Their enabling conditions differ, and their complexes can differ. Similarly, a balanced reaction combination or a qualitatively reachable product does not establish that the reactions can fire in an admissible order from a finite inventory [6]. Conversion choices can therefore preserve one analysis while changing another.

Existing software provides substantial capabilities in each area. `COBRApy` supports metabolic reconstruction and constraint-based flux analysis [7], including flux balance analysis and elementary flux modes [8, 9]. `CRNT4SBML` provides structural diagnostics and bistability analysis [10]. `MØD` combines graph rewriting with reaction-space exploration [4], and `Tellurium` and `libRoadRunner` support kinetic simulation [11, 12]. Petri-net representations of metabolic networks have a long history in structural analysis [13, 14], and Petri-net methods already provide pathway realizability certificates [6]. `SBML` and `COMBINE` facilitate model exchange [15, 16], while converters such as `KEGGtranslator` move curated pathways into that format [17]. An exchange format alone, however, does not ensure that every conversion retains the incidence and boundary assumptions needed by a particular analysis.

The remaining need is to examine conserved pools and substrate sufficiency on the same imported or generated reaction list. For example, a glycolytic route may be connected from glucose to pyruvate after ATP and redox cofactors are omitted, yet fail when their finite amounts are restored. In a generated carbohydrate network, carbon balance may hold even when a proposed sequence requires more glycolaldehyde than is available. Studying these questions together requires retaining the quantities consumed and produced, including shared participants, through each analysis. It also requires separating a substrate shortage from an inapplicable theorem or a search that stopped before reaching a conclusion.

We developed the chemical reaction network (CRN) submodule of `SynKit` [18] to address this gap. A common reaction-system object connects `KEGG` ingestion, supported `SBML` exchange, and rule-based construction to exact conservation and deficiency calculations, Petri-net diagnostics, and replayable firing certificates. The module applies established methods to the same reaction equations, so a conserved pool and an executable sequence can be interpreted under the same model assumptions.

The manuscript makes three contributions. *(i)* It describes an open-source module in which curated, exchanged, and rule-generated reactions reach exact conservation analysis, CRNT diagnostics, and finite-inventory execution checks without leaving one reaction record, and it reports the hypotheses each conclusion depends on alongside the conclusion itself. *(ii)* It validates that module against twelve reference networks, an independent `CRNT4SBML` implementation on eight shared structural quantities, and ten `BioModels` entries spanning two orders of magnitude in size, and it measures where exact enumeration becomes the operational limit. *(iii)* It uses two controlled experiments to show that representation choices that leave the net stoichiometric matrix unchanged can still change a CRNT verdict or an execution verdict: cancelling a

shared participant in glycolysis, and supplying equal carbon inventories as different precursor mixtures in a generated formose network.

2 Implementation

2.1 Reaction-network construction and representation

Within the CRN submodule, `SynCRN` stores species, reactions, rules, and provenance as reaction records. It derives a `NetworkX`-compatible bipartite digraph, stoichiometric matrices, a Petri net, and reaction equations on demand (Figure 1). The graph partitions species X and reactions $\mathscr{R}$, where species-to-reaction and reaction-to-species edges encode consumption and production. Reading each reaction and its incident edges as a directed hyperedge preserves multi-substrate dependencies. Each derived representation is calculated from these records and reflects the reaction equations at the time it is created.

For reaction r, the reactant and product multisets have coefficients s^-_{xr} and s^+_{xr}. The integer-stoichiometry analyses in this paper use

$$\mathbf{S}^- = (s^-_{xr}), \quad \mathbf{S}^+ = (s^+_{xr}), \quad \mathbf{S} = \mathbf{S}^+ - \mathbf{S}^-. \tag{1}$$

Shared participants remain in both incidence matrices. Cancelling equal coefficients preserves $\mathbf{S}$ but can change reaction enabling and the complex graph. Construction does not establish elemental balance automatically, so chemical interpretation requires suitable input equations and annotations, and the conditions under which a reaction network is chemically realizable are themselves a subject of study [19].

The `Query` adapter parses `KEGG` equations, `Construct` expands seed molecules using `SMARTS` or double-pushout graph rules, and `IO` imports supported `SBML` Core reaction structure. Reversible equations can become two directed reactions. Omitting currency metabolites, such as ATP, NADH, water, or protons, or excluding boundary species changes the analysed system. These choices are specified when constructing the network. The `Symmetry` module combines colour refinement with exact graph methods for canonicalization, isomorphism, and automorphism analysis [20, 21, 22]. Chemical labels and structural equivalence must be distinguished when interpreting these results.

2.2 Exact stoichiometric analysis and conserved pools

For dynamics $\dot{c} = \mathbf{S}v(c)$, a vector y satisfying $y^T\mathbf{S} = 0$ defines a conserved linear quantity y^Tc. `SynKit CRN` computes a rational nullspace basis and converts its vectors to primitive integer representatives. These vectors span the rational conservation space, but they need not be non-negative or form a lattice basis over $\mathbb{Z}$. Minimal P-semiflows are computed separately by exact Farkas elimination [23]: they are non-zero, non-negative integer kernel vectors with inclusion-minimal supports, normalized by their greatest common divisor. Their supports identify conserved pools of the represented network [24]. Chemical moiety identities require additional interpretation.

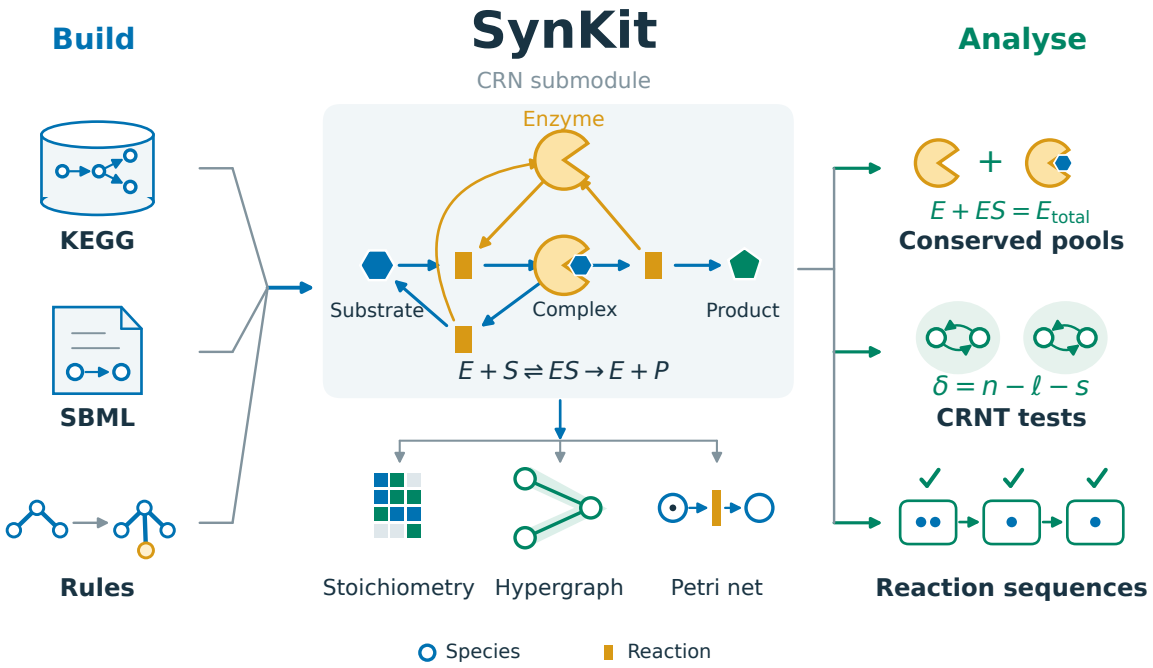


Fig. 1: Reaction representation and structural analysis workflows in `SynKit CRN`. Curated pathways (`KEGG`), models (`SBML`), and reaction rules map to a shared bipartite representation. An illustrative enzyme scheme ($E + S \rightleftharpoons ES \rightarrow E + P$, with conserved pool $E + ES$) yields stoichiometric, hypergraph, and Petri-net views for exact conservation analysis, CRNT theorem checks, and discrete pathway realizability. Molecular and diagram icons are schematic.

T-semiflows satisfy $\mathbf{S}x = 0$ with $x \geq 0$ and describe balanced reaction combinations. They do not by themselves demonstrate thermodynamic feasibility or discrete executability. Strictly positive left and right kernel vectors define conservativity and consistency, respectively. The package also exposes numerical kernel, linear-programming feasibility, and Jacobian-pattern diagnostics, which describe numerical approximations or possible matrix entries rather than exact conservation relations. A floating-point nullspace basis can accurately represent a subspace without directly identifying integer minimal supports. Naively rounding such a basis is not a substitute for exact conservation-law computation.

2.3 Network deficiency and steady-state theorem applicability

Complexes are distinct reactant or product multisets. With n complexes, ℓ connected components of the complex graph (linkage classes), and stoichiometric rank s, deficiency is

$$\delta = n - \ell - s. \tag{2}$$

A network is weakly reversible when every linkage class is strongly connected. `SynKit CRN` calculates ranks exactly over $\mathbb{Q}$ and reports global and per-class deficiencies, reversibility, and terminal strong linkage classes.

For positive mass-action rate constants, a weakly reversible deficiency-zero network has one positive steady state in each positive stoichiometric compatibility class, locally asymptotically stable relative to that class [25, 26]. A deficiency-zero network that is not weakly reversible has no positive steady state. The

Deficiency One Theorem gives at most one positive steady state per class when each linkage-class deficiency is at most one, their sum equals δ, and each linkage class has one terminal strong linkage class, while weak reversibility additionally ensures existence [27]. Failure of these hypotheses leaves the corresponding theorem undecided. Introductions to this body of theory for biologists are available [28]. Criteria that reach beyond deficiency, including injectivity, concordance, and graph-theoretic multistationarity tests [29, 30, 31], are not implemented here, and a network left undecided by the deficiency theorems is not thereby shown to be multistationary. The Edelstein example in Supplementary Figure S5 illustrates this distinction.

2.4 Petri-net analysis and discrete pathway realizability

A siphon is a non-empty species set for which every reaction producing a member also consumes a member. Once empty, it cannot refill under Petri-net firing semantics. A trap has the dual property: every reaction consuming a member also produces a member, so a marked trap cannot become empty [14]. `SynKit CRN` enumerates minimal siphons by closure-guided branch and bound [32]. This avoids inspecting many subsets but does not remove worst-case combinatorial growth.

The package checks whether every minimal siphon contains a P-semiflow support. Under the boundedness and kinetic assumptions of the corresponding persistence result, this provides a sufficient persistence certificate for positive initial conditions [33]. An uncovered siphon means the test is insufficient, and does not prove extinction or identify a unique nutrient supplementation strategy. The returned details expose these uncovered sets.

A marking m specifies the non-negative integer inventory of each species. Reaction r is enabled precisely when $m_x \geq s_{xr}^-$ for every species, and firing gives $m' = m + \mathbf{S}_{\cdot r}$. Qualitative reachability instead accumulates reachable species sets and ignores consumption and multiplicity. Given a requested flux $f \in \mathbb{N}_0^{|\mathscr{R}|}$, realizability search seeks an enabled sequence firing each reaction exactly f_r times [6]. A positive result includes a certificate: an ordered reaction list that can be checked by updating the inventory after every step. An exhaustive negative result is `unrealizable`, while reaching a state or depth limit is `inconclusive`. Certificates establish execution in the specified discrete model, without predicting concentrations, rates, or in vivo activity.

2.5 Structural `SBML` exchange and Python interface

The `SBML` Level 3 Version 2 Core adapter uses the Python standard library. It preserves supported species metadata, directed reaction stoichiometry, labels, and source identifiers on structural round trips. It does not preserve a complete kinetic model: kinetic laws, events, rules, and package-specific semantics are not supported by this adapter. Logical `SBML-qual` inputs without Core reactions are diagnosed as reaction-free. Exported `SBML` structure can be used by compatible downstream tools, with their additional kinetic or flux-bound requirements supplied separately.

Listing 1 demonstrates the public API using the cached glycolysis module. ATP/ADP/AMP, NAD(P)(H), water, phosphate, diphosphate, and protons are omitted in this example. The resulting certificate therefore

concerns a reduced model. Supplementary Section 1 describes the software architecture and additional analyses.

Listing 1: Python analysis of reduced KEGG glycolysis (M00001): network construction, exact structural analysis, discrete pathway verification, and SBML export.

```
from synkit.CRN import (crnt_summary, conserved_moieties, find_siphons,
                        siphon_persistence_details, PathwayRealizability,
                        crn_to_sbml)
from synkit.CRN.Benchmark import build_kegg_crn, GLYCOLYSIS_FLUX

# 1. Load cached KEGG pathway and remove currencies
crn = build_kegg_crn("M00001", drop_currency=True)
crn.n_species, crn.n_reactions              # (14, 15)

# 2. Structural CRNT: complexes, linkage classes, exact deficiency
report = crnt_summary(crn)
report.deficiency, report.is_weakly_reversible          # (1, False)

# 3. Exact conserved moieties, minimal siphons, structural persistence
len(conserved_moieties(crn)), len(find_siphons(crn))    # (3, 3)
siphon_persistence_details(crn).persistence_ok          # False

# 4. Discrete realizability: certify complete glycolytic flux execution
check = PathwayRealizability().load_syncrn_and_flow(
    crn, flow=GLYCOLYSIS_FLUX,
    initial_marking={"alpha-D-Glucose": 1},
    species="label", reaction="label")
check.build_petri_net_from_flow()
outcome = check.realizability_result()
outcome.status.value, len(outcome.certificate) # ("realizable", 15)
outcome.explored_states                                 # 30

# 5. Export supported reaction structure as SBML
sbml_document = crn_to_sbml(crn)
```

2.6 Validation design and computational benchmarks

Twelve reference networks provide expected structural values and regression cases. Alternative calculations check deficiency through $\mathrm{rank}(\mathbf{I}_a) - \mathrm{rank}(\mathbf{Y}\mathbf{I}_a)$, where $\mathbf{Y}$ is complex composition and $\mathbf{I}_a$ is complex-graph incidence. Numerical singular-value rank provides a further check on these small integer fixtures. Conservation vectors are checked for kernel membership, integrality, and expected count, and semiflows for non-negativity, kernel membership, and absence of nested reported supports. Exhaustive subset enumeration checks minimal siphons. These routes share input parsing and some representation helpers, so they are not wholly independent implementations, and checks on returned vectors alone do not establish enumeration completeness. An external CRNT4SBML comparison supplies an independent parser and implementation for eight shared quantities.

The larger scaling study uses three fixed network families and one timing per configuration, with hard per-task limits. A separate paired experiment repeats both siphon algorithms five times at 11, 15, 19, and 21 species, alternates execution order, uses fresh objects, excludes construction time, and records every sample plus medians and quartiles. The representation experiment keeps species identities fixed while cancelling common reactant/product coefficients in M00001, then compares inventories with and without the shared polyphosphate species. A second control retains or removes currency metabolites under the same glucose-only

marking and requested flux. Successful certificates are independently replayed against reaction records. All studies record checks and provenance in machine-readable reports.

3 Results and discussion

3.1 Reference-network validation and comparison with `CRNT4SBML`

All twelve reference networks pass their expected-value and algebraic checks (Figure 2A). The set includes enzyme mechanisms, phosphorylation cycles, the Edelstein switch, the Shinar-Feinberg motif, disconnected networks, and an open reactor [34, 35, 36]. Disconnected reversible pairs test separation of minimal conservation supports, whereas the open reactor tests inclusion of the zero complex. These are targeted correctness examples rather than a representative sample of biological networks.

`CRNT4SBML` 0.0.15 independently reads the twelve exported `SBML` networks. Species, directed reactions, complexes, linkage classes, rank, deficiency, weak reversibility, and linkage-class deficiencies agree in every case (Figure 2A). The isolated comparator environment uses Python 3.7.12, `libSBML` 5.18.0, and `NetworkX` 2.3. This comparison supports agreement on shared structural outputs, but it does not benchmark relative speed or compare bistability-search capabilities. A supplementary numerical audit finds small floating-point nullspace residuals on all twelve fixtures, while naive scaling and integer rounding produces invalid vectors on 3 networks (Supplementary Table S7). Exact integer vectors satisfy the identity in every case. This supports exact reconstruction for integer invariants without claiming that numerical subspace estimation is intrinsically inaccurate.

3.2 Computational cost of exact structural enumeration

Figure 3A reports the repeated paired experiment. Both algorithms return identical minimal siphons for every run. At 21 species, median exhaustive-search time is 3.01 s and median closure-search time is 0.84 ms. The median ratio between the two rises from roughly 12 at 11 species to roughly 3600 at 21 species, so the two methods separate over this range rather than differing by a constant factor. Quartiles describe repeat-to-repeat variation on these fixed networks, rather than variation across network topologies, and four sizes on one network family do not determine how the ratio continues.

The larger archived sweep includes directed chains, reversible chains, and random sparse bimolecular networks (Supplementary Figure S6). On the 401-species reversible chain, minimal siphons take 8.4 s, canonicalization 14.1 s, and exact conserved moieties 228.6 s (Figure 3B). On the random family, moiety computation exceeds 300 s at 197 species, so the next size is omitted for that task. Siphons complete at 395 species in 15.8 s. These single-run results locate operational limits on the recorded machine. Neither smooth curves nor results for three families establish polynomial scaling or a universal species-count ceiling.

A **Reference checks and independent SBML comparison**

	Species	Rxns	n	ℓ	s	δ	Laws	WR	Tests	External
Reversible isomerization	2	2	2	1	1	0	1	●	✓	8/8
Irreversible isomerization	2	1	2	1	1	0	1	○	✓	8/8
Isomerization cycle	3	3	3	1	2	0	1	●	✓	8/8
Disconnected reversible pairs	6	6	6	3	3	0	3	●	✓	8/8
Pairs with stoichiometry	6	6	6	3	3	0	3	●	✓	8/8
Michaelis–Menten	4	3	3	1	2	0	2	○	✓	8/8
Reversible Michaelis–Menten	4	4	3	1	2	0	2	●	✓	8/8
Edelstein switch	3	6	5	2	2	1	1	●	✓	8/8
Phosphorylation: one site	6	6	6	2	3	1	3	○	✓	8/8
Phosphorylation: two sites	9	12	10	2	6	2	3	○	✓	8/8
Shinar–Feinberg motif	2	2	4	2	1	1	1	○	✓	8/8
Open reactor	2	3	3	1	2	0	0	●	✓	8/8

Colour scaled within each count column; WR: ● yes / ○ no; external: shared quantities agreeing.

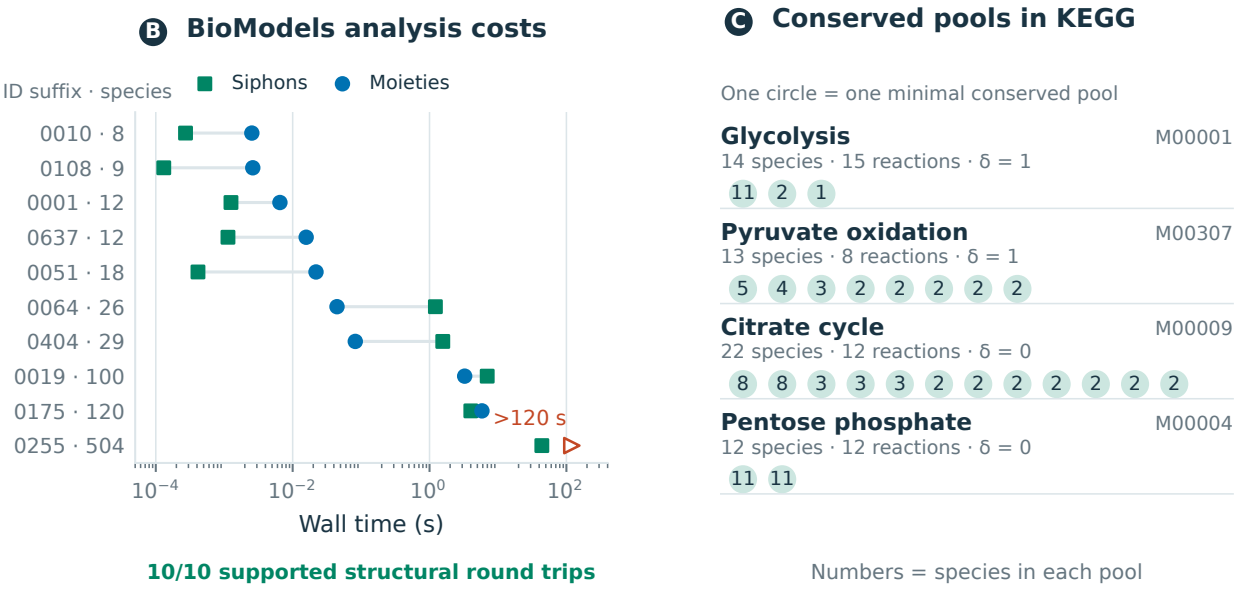


Fig. 2: Structural validation, `BioModels` scaling, and conserved pools in `KEGG` modules. **(A)** Structural invariants of twelve reference networks (n, complexes, ℓ, linkage classes, s, stoichiometric rank, δ, deficiency, laws, conservation dimension). Filled and open circles denote weakly reversible (WR) and non-weakly-reversible networks. Tests indicates internal consistency, while External indicates agreement with `CRNT4SBML` across eight structural metrics. **(B)** Enumeration runtimes for minimal siphons and conserved pools across ten `BioModels` entries ordered by species count (open triangle: 120 s timeout for moiety enumeration). **(C)** Minimal non-negative conserved pools in four reduced `KEGG` modules. Circles denote individual pools labelled by species support size, and annotations report network dimensions and deficiency.

3.3 Structural `SBML` round-trip validation in `BioModels`

A size-stratified manifest selects eleven `BioModels` entries [37]: ten reaction-bearing models and an `SBML-qual` negative control. All ten pass the supported structural export/re-import checks, and their deficiency identities hold (Figure 2B). Source identifiers, URLs, and input checksums accompany the results. The largest, `BIOMD0000000255`, contains 504 species and 827 directed reactions. Minimal siphons take 46.7 s, while exact moiety enumeration exceeds its 120-second budget. This limit prevents a complete persistence assessment for that model. `BIOMD0000000562` is correctly diagnosed as reaction-free for Core structural analysis. The selected cohort tests interoperability over a range of sizes, but does not estimate repository-wide support.

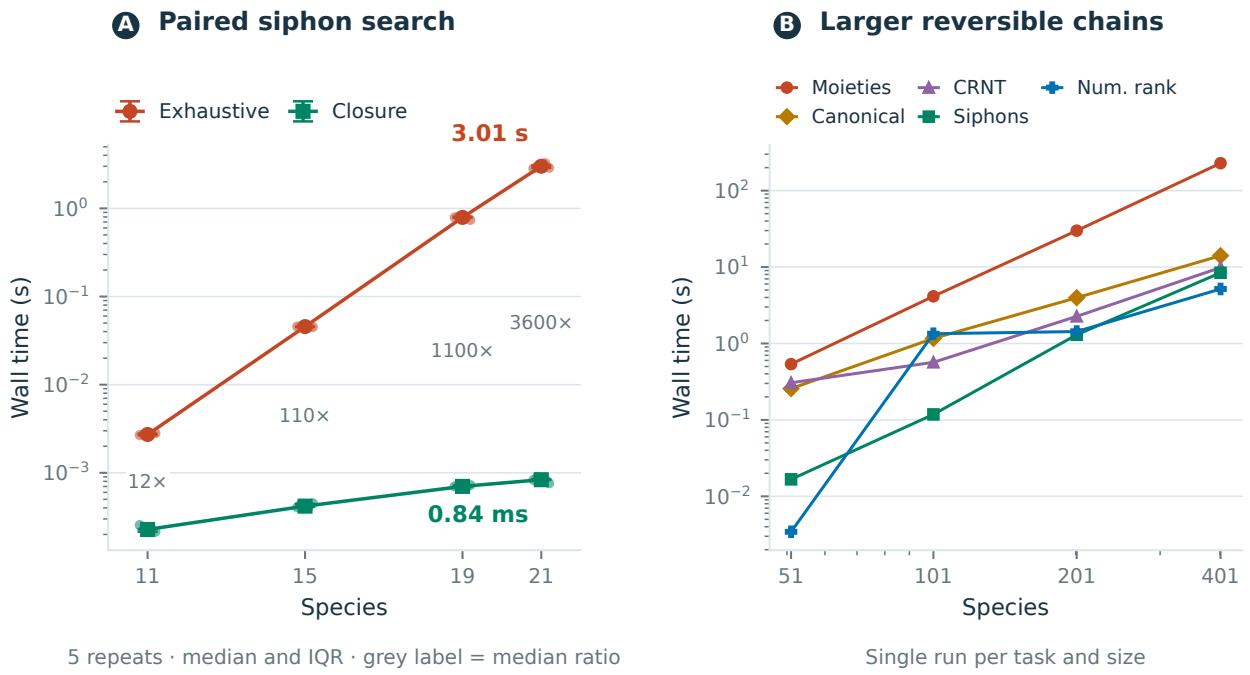


Fig. 3: Runtime comparison of siphon algorithms and structural analyses on reversible-chain networks. **(A)** Exhaustive versus closure-guided minimal-siphon enumeration across five repeats per network size (points show individual runs, markers and error bars indicate medians and IQRs, and both methods yield identical siphon sets). Grey labels give median runtime ratios. **(B)** Single-run execution times for five structural analyses across larger reversible chains (logarithmic scale, with lines as guides to the eye).

3.4 Conservation and cofactor requirements in glycolysis

Four cached `KEGG` modules cover glycolysis (`M00001`), pyruvate oxidation (`M00307`), the citrate cycle (`M00009`), and the pentose phosphate pathway (`M00004`) [1]. Their conserved-pool supports appear in Figure 2C. In the reduced citrate-cycle model, minimal non-negative conservation vectors recover nucleotide pairs, a quinone pair, a ferredoxin pair, coenzyme A, and a lipoyl pool. These interpretations follow from the retained reaction equations and labels, so they do not establish organism-specific physiological pool conservation.

The reduced glycolysis model contains 14 species and 15 directed reactions, with $n = 16$, $\ell = 4$, $s = 11$, and $\delta = 1$. Its largest minimal conservation support spans eleven carbohydrate intermediates, assigning weight two to hexoses and one to trioses (Figure 4). Glucose and oxidized ferredoxin form uncovered singleton siphons. Polyphosphate forms a covered singleton because the cached generic-polymer reaction retains it on both sides. This symbolic representation does not resolve polymer chain length or prove catalytic phosphate chemistry.

For one glucose token, the prescribed flux has a 15-step firing certificate ending in two pyruvate tokens, and search explores 30 states. The repeated lower-pathway firings are specified in the requested flux and are compatible with the two trioses produced by aldolase. Certificate replay verifies enabling at each step and conservation of the weighted carbohydrate pool. Requesting only the lower pathway from glucose fails because the required intermediates are absent.

The controlled experiment isolates the information lost by cancellation (Figure 5). Removing the common polyphosphate incidence from `R02189` preserves every species row, reaction column, and entry of $\mathbf{S}$.

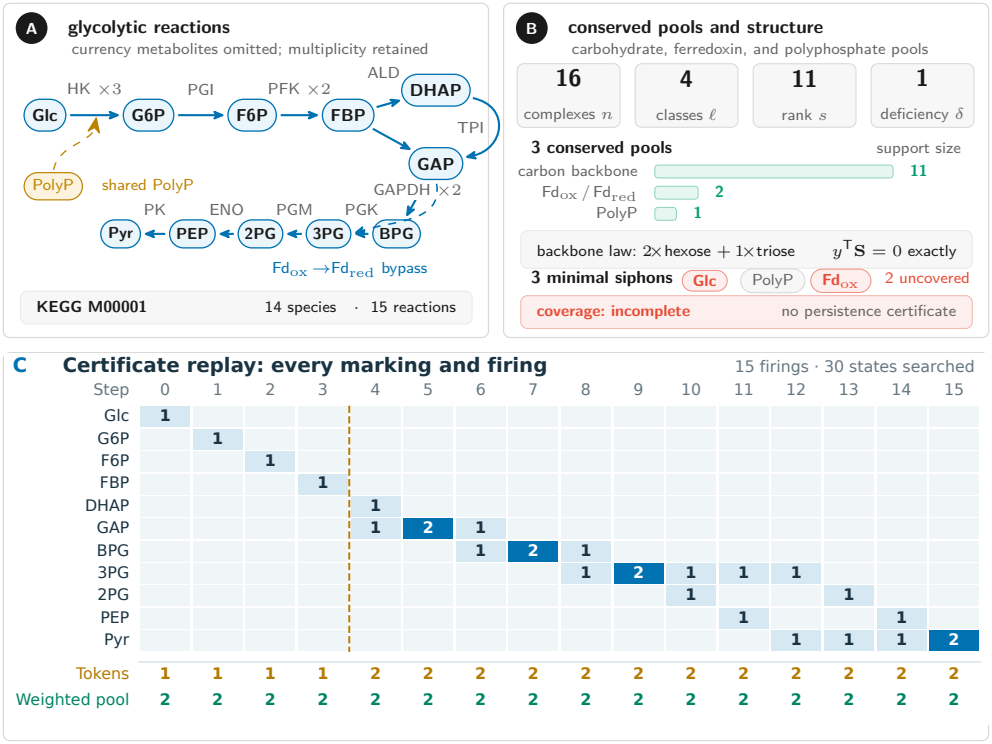


Fig. 4: Conserved pools and discrete pathway execution in reduced KEGG glycolysis (M00001). **(A)** Bipartite network (14 species, 15 directed reactions) retaining shared polyphosphate and the ferredoxin bypass, where edge labels indicate parallel alternatives. **(B)** Structural invariants, conserved-pool support sizes, and minimal siphons. Singleton siphons (glucose, oxidized ferredoxin) lack semiflow supports, leaving persistence uncertified. **(C)** Token markings across the certified 15-step conversion of one glucose to two pyruvates (blank cells denote zero, orange rule indicates aldolase cleavage). Bottom rows trace total carbohydrate tokens and the conserved weighted pool ($2 \times \text{hexoses} + 1 \times \text{trioses} = 2$). Inactive polyphosphate and ferredoxin are omitted.

Consequently, its kernel and conservation space are unchanged. Nevertheless, the complex count changes from 16 to 14, linkage classes from four to three, and deficiency from one to zero. With one glucose and zero polyphosphate tokens, the intact reaction is blocked, whereas the cancelled reaction executes. Supplying one polyphosphate token enables the intact reaction. Thus net-stoichiometric equivalence does not imply CRNT or execution equivalence.

A separate boundary control restores the currency metabolites, producing a 24-species model with the same 15 reactions. Under the same glucose-only inventory and requested flux, no first firing is enabled and exhaustive search returns `unrealizable`. The 15-step certificate therefore depends on currency removal. The reduced sequence establishes conversion of the carbohydrate intermediates, and evaluating cofactor sufficiency requires restoring the omitted species and specifying their initial amounts. Supplementary Section 2.1 gives a cofactor-coupled example with explicit ATP and ADP alternatives.

3.5 Carbon conservation and precursor requirements in formose networks

The formose reaction, in which formaldehyde condenses to a mixture of sugars [38], gives a rule-generated counterpart to the curated KEGG modules. Starting from formaldehyde and glycolaldehyde, four rules for aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization generate 37 species and 44 reactions after four iterations. Generation takes 0.28 s in the archived run and reaches species containing

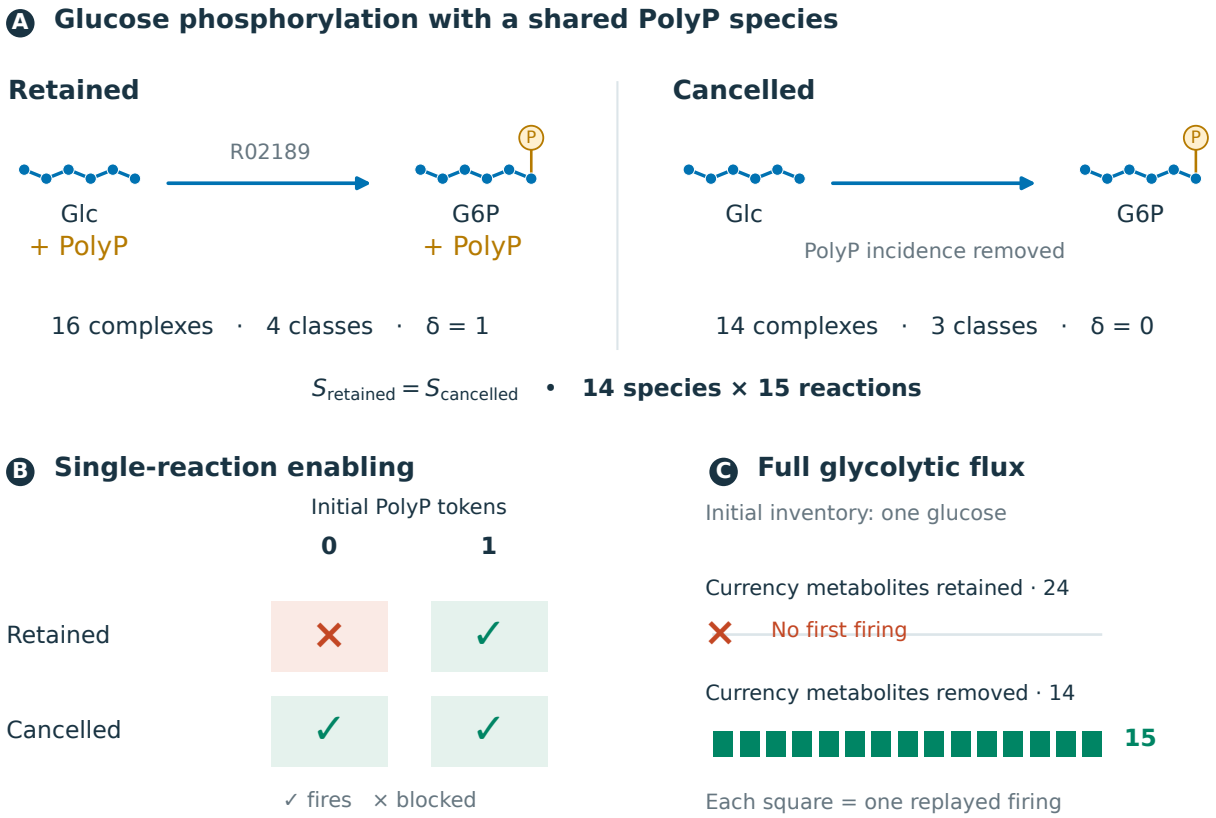


Fig. 5: Sensitivity of structural invariants and pathway realizability to participant retention and currency omission. **(A)** Cancelling shared polyphosphate (`PolyP`) in reaction `R02189` preserves net stoichiometry (**S**) but changes complex count (n), linkage classes (ℓ), and deficiency (δ). Glc indicates glucose, G6P indicates glucose 6-phosphate, blue circles represent carbon count, and P represents phosphate. **(B)** Enabling of `R02189` from one glucose under zero versus one `PolyP` token. **(C)** Realizability of the prescribed glycolytic flux from glucose alone. Retaining currency metabolites blocks execution, whereas omitting them permits a certified 15-step sequence (ticks show verified firing certificates, and crosses indicate exhaustive failure).

ten carbon atoms. The iteration count is a rule-expansion depth, not physical reaction time. The network has 54 complexes, 17 linkage classes, rank 36, and deficiency one.

Its one-dimensional left nullspace yields a positive primitive conservation vector. Comparing its coefficients with carbon counts from the molecular graphs gives exact agreement for all 37 species (Figure 6). This is a consistency check between generated stoichiometry and molecular structures. Within this carbohydrate-only rule set, elemental counts can be proportional, but the result does not show that stoichiometry uniquely identifies elemental composition in general.

Qualitative search reaches representative tetrose and hexose targets from the two seed species, but discrete execution distinguishes inventories that share the same total carbon (Figure 6D). With four formaldehyde and one glycolaldehyde molecule, neither selected route executes. With two of each precursor, the tetrose route executes but the hexose route does not. With three glycolaldehyde molecules, both execute. Each of these inventories contains six carbon atoms. The selected routes therefore require suitable precursor multiplicities as well as a sufficient carbon budget. These results concern fixed reaction multisets in the generated network, and they do not establish unrestricted target unreachability or experimental synthetic feasibility. Supplementary Section 2.2 gives the route definitions and initial pools.

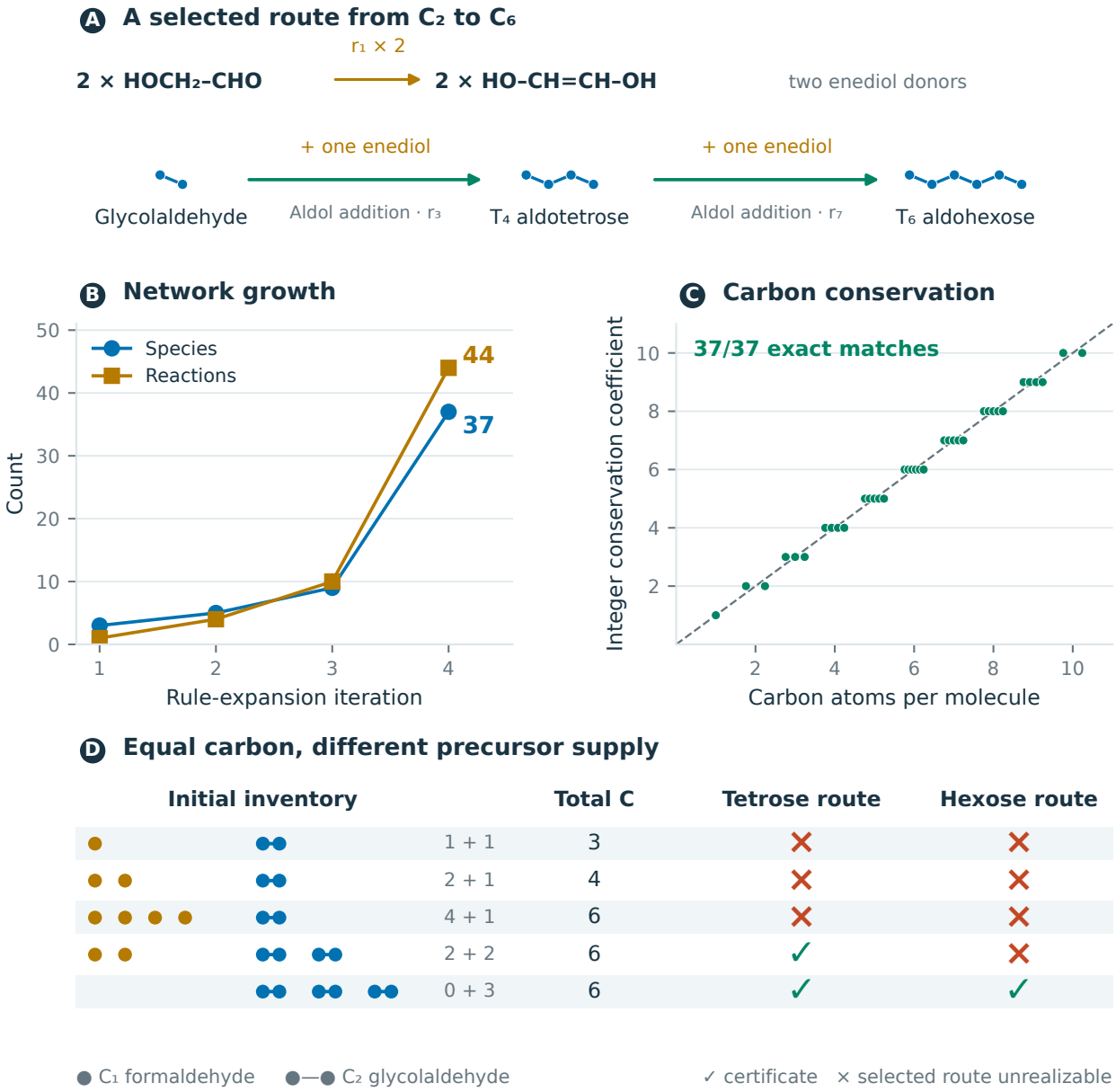


Fig. 6: Carbon conservation and precursor-dependent pathway realizability in the generated formose network. **(A)** Target hexose route requiring three glycolaldehyde precursors via two enolizations and two aldol additions (blue circles: carbon count). **(B)** Network growth across four rule-expansion iterations from formaldehyde and glycolaldehyde. **(C)** Exact agreement between primitive stoichiometric conservation coefficients and molecular carbon counts for all 37 species (points jittered horizontally, dashed line: $y = x$). **(D)** Realizability of selected tetrose (T_4) and hexose (T_6) routes across five initial inventories (orange dots represent formaldehyde, blue pairs represent glycolaldehyde, ticks show verified certificates, and crosses indicate exhaustive failure). Inventories of identical total carbon differ in their ability to supply these routes.

3.6 Biochemical interpretation of conservation and pathway realizability

A conserved pool describes quantities that reactions redistribute, whereas an execution check asks whether the required substrates are available at every step. These questions remain distinct in both case studies. Glycolysis conserves its weighted carbohydrate pool even when removal of a shared participant changes reaction requirements and deficiency. In formose, six carbon atoms supplied as different precursor mixtures support different selected routes. Thus conservation is necessary for interpreting these conversions but does not establish that a particular substrate inventory can carry them out.

3.7 Comparison with existing reaction-network software

The software comparison concerns the CRN submodule of `SynKit`. Its underlying methods are established and are also available in other packages. What the module adds is that one reaction record reaches all of them, so a conserved pool, a deficiency verdict, and a firing certificate refer to the same species inventory, the same treatment of shared participants, and the same boundary choices. Table 1 places this scope alongside the packages closest to the analyses in this paper.

Table 1: Functional scope of biochemical reaction-network software (✓ documented, (✓) partial or indirect as detailed in footnotes, × not documented). Feature presence is derived from package documentation and source code, and only `CRNT4SBML` was experimentally evaluated (Figure 2A). The `SynKit` column describes its CRN submodule.

	`SynKit` (this work)	`CRNT4SBML` [10]	`CoNtRol` [39]	`crnpy` [40]	`COBRApy` [7]	`MØD` [4]	`Tellurium` [11, 12]
Network construction and exchange							
`KEGG` pathway ingestion	✓	×	×	×	×	×	×
`SBML` import	(✓)[a]	✓	×	✓	✓	×	✓
`SBML` export	(✓)[a]	×	×	✓	✓	×	✓
Rule-based generation from seeds	✓	×	×	×	×	✓	×
Exact structural analysis							
Deficiency, linkage classes, weak reversibility	✓	✓	✓	✓	×	×	×
Conservation laws (kernel basis)	✓	✓	×	✓	×	×	✓
Minimal P-semiflows (minimal support)	✓	×	×	✓	×	×	×
T-semiflows or elementary modes	✓	×	×	✓	×	×	×
Minimal siphons and traps	✓	×	(✓)[b]	×	×	×	×
Behaviour and execution							
Multistationarity or bistability search	×	✓	✓	(✓)[c]	×	×	×
Finite-inventory firing certificate	✓	×	×	×	×	(✓)[d]	×
Constraint-based flux analysis	×	×	×	×	✓	×	×
Kinetic simulation	×	(✓)[e]	×	×	×	×	✓

[a] `SBML` Level 3 Version 2 Core reaction structure and metadata only, as kinetic laws, events, rules, and package extensions are not handled.
[b] Siphons underlie `CoNtRol`'s persistence conditions, but minimal-siphon enumeration is not exposed as an output.
[c] `crnpy` supplies Gröbner-basis, elementary-mode, and absolute-concentration-robustness machinery rather than a bistability search routine.
[d] `MØD` provides an integer hyperflow model for pathway queries rather than a firing-sequence certificate over a finite marking.
[e] ODE simulation is provided in support of `CRNT4SBML`'s bistability analysis.

Read across, the table says where the module is and is not the right tool. The CRN submodule of `SynKit` combines rule-based construction, minimal siphons and traps, and finite-inventory firing certificates in one object, and only `crnpy` matches its exact semiflow analysis. It has no bistability search, no constraint-based flux analysis, and no kinetic simulation, so `CRNT4SBML`, `CoNtRol`, `COBRApy`, and `Tellurium` remain the appropriate tools for those questions, and a complete study may well use two packages. Our external comparison establishes agreement on shared CRNT outputs and nothing beyond that: we have not compared usability, total runtime of complete analyses, or how deeply each package develops a capability the table marks present. The case for the module rests on the controlled experiments of Sections 3.4 and 3.5, in which the answer changes depending on how the shared reaction record is prepared. That failure mode is only visible when the analyses share a representation.

3.8 Model assumptions and computational limitations

The conclusions depend on the reaction list, cofactor treatment, reversibility, and boundary species. The glycolysis controls quantify two consequences of changing these assumptions. Minimal-semiflow and siphon enumeration can have combinatorial output and runtime growth. Timeouts and incomplete search must remain visible. Structural identities and certificates do not predict kinetic rates, thermodynamic driving forces, or in vivo activity. Theorem conclusions require their hypotheses, and supported `SBML` round trips do not preserve a complete simulation model. Future work should prioritize broader input validation and scalable exact enumeration with explicit resource reporting.

4 Conclusions

The CRN submodule of `SynKit` connects curated and rule-generated reactions to conservation analysis, CRNT diagnostics, and discrete pathway certificates. Reference checks and an external `SBML` comparison support its structural computations, while metabolic and formose examples show how to interpret the outputs. The controlled glycolysis experiment demonstrates why preserving reactant and product incidence matters even when net stoichiometry is unchanged. Together, the analyses identify conserved metabolite pools and test whether specified substrate amounts support a proposed reaction sequence. Their interpretation remains conditional on the biochemical species and reactions included in the model.

Declarations

Ethics approval and consent to participate

Not applicable.

Availability of data and materials

The CRN submodule is distributed with `SynKit`. Source code is available on GitHub at github.com/TieuLongPhan/SynKit, installation is available via PyPI at pypi.org/project/synkit/, documentation is hosted at tieulongphan.github.io/SynKit/, and the public v1.6.2 release is archived on Zenodo [41]. `SynKit` is released under the MIT License. The accompanying reproducibility documentation identifies the source revision, subsequent changes, and input data used for the analyses. The archived release and the implementation used in this manuscript are distinguished explicitly. The `KEGG` module records underlying Figure 2C are derived from `KEGG` [1] and cached for reproducibility, so they are not covered by `SynKit`'s MIT licence and remain subject to `KEGG`'s terms. The `BioModels` inputs are identified and checksum-recorded in the evidence, while `BioModels` distributes its dataset under the CC0 1.0 Public Domain Dedication.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Author contributions are reported using the CRediT taxonomy. **T.-A.P.**: Software, Validation, Writing (original draft), Writing (review and editing). **T.-L.P.**: Conceptualization, Methodology, Software, Validation, Writing (original draft), Writing (review and editing). Both authors read and approved the final manuscript.

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List of abbreviations

ACR: absolute concentration robustness. CRN: chemical reaction network. CRNT: chemical reaction network theory. DPO: double pushout. EMP: Embden-Meyerhof-Parnas. `KEGG`: Kyoto Encyclopedia of Genes and Genomes. `SBML`: Systems Biology Markup Language. `SMARTS`: SMILES arbitrary target specification. TCA: tricarboxylic acid. WR: weakly reversible.

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